

```

1  Program Question_1
2  Integer,Dimension(4,4)::matA
3  Integer::i,j,d2,d3
4  Open(unit=23,file='inp1.txt')
5  Open(unit=27,file='out1.txt')
6  !For Input
7  Do i=1,4
8      Do j=1,4
9          Read(23,*)matA(i,j)
10         End Do
11     End Do
12     !output for (i)
13     write(27,*)"Matrix A"
14     Write(27,'(4I3)') ((matA(i,j),j=1,4),i=1,4)
15
16     !output for (ii)
17     write(27,*)"Matrix A in Horizontal line traversing row-wise"
18     Write(27,'(//16I3//)') ((matA(i,j),j=1,4),i=1,4)
19
20     !output for (iii)
21     write(27,*)"Matrix A in a vertical line"
22     Write(27,'(I3)') ((matA(i,j),i=1,4),j=1,4)
23
24     !output for iv
25     Write(27,*)"Only the elements of 1st row and 1st column"
26     write(27,'(//4I3)') (matA(1,j),j=1,4)
27     write(27,'(I3)') (matA(i,1),i=2,4)
28
29     !output for (v)
30     Write(27,'(//A)') "Result of v"
31
32     d3=matA(1,1)**3+matA(2,2)**3+matA(3,3)**3+matA(4,4)**3
33     d2=matA(1,4)**2+matA(2,3)**2+matA(3,2)**2+matA(4,1)**2
34     Write(27,*)"d2=",d2,"d3=",d3
35     If(d2>d3)then
36         Write(27,*)"d2 is large "
37     else
38         Write(27,*)"d3 is large"
39     End If
40
41     !output for vi
42     Write(27,'(//A//)') "Transpose of A"
43     Write(27,'(4I3)') ((matA(j,i),j=1,4),i=1,4)
44
45
46
47  End Program
48

```